

Some Other Simple and Precise Quantitative equations in Protein Structure Prediction

Kequn Li

(School of Chemical Science and Engineering, Tongji University,

Shanghai, 200092)

Abstract: In studies conducted on three-dimensional coordinate data of proteins downloaded from the world-renowned Protein Data Bank (PDB), it was discovered that the force balance problem of the main chain of many proteins can be divided into two parts. The first part is the backbone portion, which is the subject of the author's previous proposal: assuming a unidirectional tensile force along the protein main chain from one end to the other, the sum of the projections of the tensile forces between two adjacent central atoms along the X , Y and Z -axes equals zero, thus satisfying the force balance conditions. The second part is the non-backbone portion. When forces are directed from the central atoms of the protein main chain to their associated non-central atoms, if all forces are assumed equal, the sum of their projections along the X , Y and Z -axes also equals zero, indicating a state of force balance. Although the force balance of the protein side-chain backbone portion is still under investigation, other part other than the side-chain backbone portion also show good agreement with the force balance conditions. These simple and precise quantitative relationships can greatly facilitate research on protein structures. However, it should be noted that not all proteins satisfy these precise relationships. We believe that deviations from the force balance conditions are related to factors such as the binding of small organic molecules to proteins, which will be further explored in subsequent work.

Keywords: Protein structure prediction; Force balance; Main chain; Side chain; Quantitative equation

Proteins are one of the important biological macromolecules, serving as the material basis of life, essential components of organisms, and major executors of life activities. Proteins perform various vital functions in organisms, such as catalysis of biochemical reactions, transport of substances, control of growth and differentiation, and signal recognition and transduction. Biological processes such as reproduction, development, and growth also depend on proteins. The ability of proteins to perform important functions in life activities depends on their highly complex three-dimensional structures¹. Protein structure is crucial for biological research, disease treatment, and even industrial applications². Traditional experimental methods, including X-ray crystallography, nuclear magnetic resonance imaging, and cryo-electron microscopy, are time-consuming and expensive³⁻⁴, limiting their practicality in large-scale applications. As a result, the number of experimentally determined protein structures falls far short of the demands of modern science. Therefore, there has been a persistent desire to predict protein three-dimensional structures through computational methods. However, traditional protein structure prediction algorithms, primarily based on homology modeling of known protein

structures and prior physical knowledge, have low prediction accuracy and deviate significantly from experimentally determined data⁴. Moreover, even the currently most prominent model, AlphaFold from Google, which employs AI technology and demonstrates significantly higher accuracy than most other prediction methods⁵, has been argued by some biologists as a "black-box" model with insufficient physical interpretability and mechanistic clarity⁶⁻⁷. Like other existing prediction methods, it does not truly address the theory or mechanism of protein folding⁸⁻¹⁰. The author previously proposed that, assuming the backbone of a single-chain protein main chain (i.e., the long chain of amino nitrogen, α -carbon, and carbonyl carbon atoms) is subjected to a uniform tensile force in a single direction, the protein backbone constitutes a "force-balanced but torque-imbalanced system"¹¹, thereby revealing simple and precise physical relationships within protein structures. This paper continues to introduce other simple and precise quantitative equations discovered from single-chain protein data downloaded from the world-renowned Protein Data Bank (PDB), with the expectation of revealing more underlying physical principles of protein structures, enabling deeper understanding and research in the future.

1. Theoretical Section

1.1 Force Balance Problem in Cable-Membrane Structures

Cable-membrane structures, composed of cables and membranes, are common in civil engineering. A cable-membrane structural unit is a system where force balance is achieved through tensile forces of cables connecting a central cable node and several points on the membrane. For convenience in studying protein systems, we define the direction of the cable tension as pointing from the central cable node towards the respective membrane point.

If the three-dimensional coordinates of the central cable node O are (x_o, y_o, z_o) , and the three-dimensional coordinates of the points on the membrane are (x_i, y_i, z_i) , then the projections of the tensile forces f_i from each cable along the x -axis are (for convenience of description, only the x -axis is discussed here; the y - and z -axis cases can be deduced similarly):

$$F_{xi} = f_i \frac{x_i - x_o}{\sqrt{(x_i - x_o)^2 + (y_i - y_o)^2 + (z_i - z_o)^2}} \quad (1)$$

The sum of the projections of the cable tensile forces along the x -axis is:

$$F_x = \sum_{i=1}^m f_i \frac{x_i - x_o}{\sqrt{(x_i - x_o)^2 + (y_i - y_o)^2 + (z_i - z_o)^2}} \quad (2)$$

In equation 2, m is the total number of cables in the cable-membrane structural unit. Similarly, we can derive the sums of the projections of cable tensile forces along the y - and z -axes. According to engineering mechanics principles, if the sums of the tensile forces along the x -, y -, and z -axes are zero, the cable-membrane structural unit is considered a force-balanced system.

2. Minimum Total Cable Length and Minimal Surface Problem of Cable-Membrane Structures

It is easy to understand that the sum of the lengths of all cables l_a in a cable-membrane structural unit can be expressed as:

$$l_a = \sum_{i=1}^m \sqrt{(x_i - x_o)^2 + (y_i - y_o)^2 + (z_i - z_o)^2} \quad (3)$$

According to calculus, the condition for minimizing the total cable length can be obtained by taking the first partial derivatives of l_a with respect to x_o , y_o ,

and z_o and setting them to zero. Taking the x -axis as an example:

$$\frac{\partial l_a}{\partial x_o} = -\sum_{i=1}^m \frac{x_i - x_o}{\sqrt{(x_i - x_o)^2 + (y_i - y_o)^2 + (z_i - z_o)^2}} = 0 \quad (4)$$

It can be seen that if the tensile forces f_i in equation 2 are equal, then equation 4 is

equivalent to equation 2 being zero. The y - and z -axis cases are analogous. In this

case, the sum of cable lengths in the cable-membrane structure is minimized.

Reference 12 points out that the membrane in such a cable-membrane structural unit is a minimal surface.

3. Minimum Length and Force Balance Problem of Long Chains in Protein Structures

If we connect many cable-membrane structural units by sharing cables, we form a long-chain structure similar to a protein structure, where both the main chain and side

chains can be considered as composed of such long chains. Taking the protein main chain as an example, it can be divided into two parts as a backbone portion (the long chain formed by central atoms on the main chain, such as amino nitrogen, α -carbon, and carbonyl carbon atoms) and a non-backbone portion (small branches on the main chain formed by these central atoms binding to respective non-central atoms, such as hydrogen atoms, β -carbon atoms, or carbonyl oxygen atoms). The total length l of the protein main chain is the sum of these two parts:

$$l = \sum_{i=1}^n \sqrt{(x_{o,i} - x_{o,i+1})^2 + (y_{o,i} - y_{o,i+1})^2 + (z_{o,i} - z_{o,i+1})^2} + \sum_{i=1}^{n+1} \sum_{j=1}^{p(i)} \sqrt{(x_{ij} - x_{o,i})^2 + (y_{ij} - y_{o,i})^2 + (z_{ij} - z_{o,i})^2} \quad (5)$$

In equation 5, $x_{o,i}$ and x_{ij} are the x -coordinates of the i -th central atom on the protein main-chain backbone and the j -th non-central atom bound to it, respectively; $p(i)$ is the number of non-central atoms bound to the i -th central atom (e.g., for α -carbon, it is generally 2).

To determine the conditions for minimizing the total length l , we take the first partial derivatives with respect to the x -coordinates of each central atom on the protein main-chain backbone and set them to zero. Taking the i -th central atom on the backbone as an example, partial differentiation on the total length l with respect to $x_{o,i}$ only affects the terms in equation 5 involving $x_{o,i}$, and the solution process is the same as in section 1.2 for cable-membrane units. According to calculus, when every central atom on the protein main-chain backbone is in force balance and the forces between any two bonded atoms in the main chain are equal (notice the mutual tensile forces between two adjacent central atoms are internal forces, equal in magnitude and opposite in direction), the total length of the protein main chain is minimized. In civil engineering, this is referred to as the surface formed by all non-central atoms on the protein main chain being a minimal surface¹³⁻¹⁴. We found that protein molecules perfectly satisfying these conditions are uncommon. However, if we sum the first partial derivative expressions of the total length l (from equation 5) with respect to the three-dimensional coordinates of all central atoms, we obtain more general expressions commonly encountered in practical work. Still using the x -axis as an example, since the total length is the sum of bond lengths between pairs of bonded atoms, we also divide these expressions into two parts:

(a) Protein Backbone Portion

Taking the i -th and $i+1$ -th adjacent central atoms on the protein main chain as an example, the length between them is:

$$l_1 = \sqrt{(x_{o,i} - x_{o,i+1})^2 + (y_{o,i} - y_{o,i+1})^2 + (z_{o,i} - z_{o,i+1})^2} \quad (6)$$

The first partial derivative of equation 6 with respect to $x_{o,i}$ is:

$$\frac{\partial l_1}{\partial x_{o,i}} = \frac{x_{o,i} - x_{o,i+1}}{\sqrt{(x_{o,i} - x_{o,i+1})^2 + (y_{o,i} - y_{o,i+1})^2 + (z_{o,i} - z_{o,i+1})^2}} \quad (7)$$

The first partial derivative of equation 6 with respect to $x_{o,i+1}$ is:

$$\frac{\partial l_1}{\partial x_{o,i+1}} = -\frac{x_{o,i} - x_{o,i+1}}{\sqrt{(x_{o,i} - x_{o,i+1})^2 + (y_{o,i} - y_{o,i+1})^2 + (z_{o,i} - z_{o,i+1})^2}} \quad (8)$$

From equations 7 and 8, it can be seen that these two parts cancel each other out in the first partial derivative of equation 6. Similarly, this can be extended to other adjacent central atoms on the protein backbone; their first partial derivatives will cancel each other, so this part can be ignored in calculations.

(b) Protein Main Chain Non-Backbone Portion

From equation 5, taking the i -th central atom on the protein main chain as an example, the sum of the first partial derivatives with respect to $x_{o,i}$ of the lengths of the small branches formed by the bonds between this central atom and its non-central atoms can be expressed as:

$$\Delta_{x,i} = -\sum_{j=1}^{p(i)} \frac{x_{ij} - x_{o,i}}{\sqrt{(x_{ij} - x_{o,i})^2 + (y_{ij} - y_{o,i})^2 + (z_{ij} - z_{o,i})^2}} \quad (9)$$

If we sum the corresponding $\Delta_{x,i}$ for each central atom on the protein main chain and set the sum to zero, we obtain a force balance expression:

$$\Delta_x = \sum_{i=1}^{n+1} \sum_{j=1}^{p(i)} \frac{x_{ij} - x_{o,i}}{\sqrt{(x_{ij} - x_{o,i})^2 + (y_{ij} - y_{o,i})^2 + (z_{ij} - z_{o,i})^2}} = 0 \quad (10)$$

It is easy to see that this expression actually indicates that the non-backbone portion of the protein main chain is in a state of force balance. Considering experimental errors in protein structure determination, we convert equation 10 into a proportional form:

$$r_x = \frac{\sum_{i=1}^{n+1} \sum_{j=1}^{p(i)} \frac{x_{ij}}{\sqrt{(x_{ij} - x_{o,i})^2 + (y_{ij} - y_{o,i})^2 + (z_{ij} - z_{o,i})^2}}}{\sum_{i=1}^{n+1} \sum_{j=1}^{p(i)} \frac{x_{o,i}}{\sqrt{(x_{ij} - x_{o,i})^2 + (y_{ij} - y_{o,i})^2 + (z_{ij} - z_{o,i})^2}}} = 1 \quad (11)$$

The y - and z -axis cases can be deduced similarly. Considering that reference 11 points out that for the backbone portion of many protein main chains, assuming a unidirectional tensile force from one end to the other with equal forces between adjacent central atoms, the sum of the projections of these forces along the x -, y - and z -axes is zero, which means the backbone portion is also in force balance. Therefore, for protein main chains satisfying both force balance conditions (backbone and non-backbone portions), they can be studied separately from the perspective of force balance, greatly facilitating related work.

Furthermore, much or less, since most proteins have some central atoms that satisfy the force balance conditions, we hypothesize that there may be complementary substructures within these protein main chains, where deviations from the force balance condition by certain central atoms may be offset by others, ultimately satisfying the expression in equation 11.

As shown in equations 7 and 8, the first partial derivatives of the length between two adjacent central atoms on the protein main chain with respect to their three-dimensional coordinates cancel each other out. This principle can be extended to side chains containing atoms linked to three central atoms, such as the β -carbon in valine which is connected to three carbon atoms. It is understandable that in force balance calculations for protein side chains, only the forces between side-chain central atoms and side-chain non-central atoms need to be considered.

2. Calculation Results and Discussion

2.1 Protein Main Chain Portion

The experimentally determined protein three-dimensional coordinate data used in this study were obtained from the world-renowned Protein Data Bank (PDB) at www.rcsb.org. By substituting the relevant atomic three-dimensional coordinate data from downloaded protein structure CIF files into equations 10 and 11, the force balance calculation results for the non-backbone portion of the protein main chain are shown in Table 1. In the calculations, the interaction between the α -carbon and the side chain is considered only via the bond to the β -carbon atom.

Table 1. Force balance calculation results for the non-backbone portion of some protein main chains

No.	Species	Protein Name	PDB ID	Δ_x	Δ_y	Δ_z	r_x	r_y	r_z
1	Sperm whale	Myoglobin	1CQ2	-62.00	11.70	3.58	0.99607	1.0005	1.00038
2	<i>Homo sapiens</i>	Plasminogen activator	1KDU	162.20	-4.12	-0.11	1.0086	0.998680	0.999980
3	<i>Bacillus licheniformis</i>	KerA protein	6DWQ	77.70	6.30	-4.40	1.00115	0.999854	0.999618
4	<i>Rhodobacter capsulatus</i>	Cytochrome C	1CPQ	4.80	-48.00	-82.10	1.00021	0.997484	0.996469

Simultaneously, following the method in reference 11, we calculated the force balance results for the central atoms of the main chain backbone for the above proteins, as shown in Table 2.

Table 2. Force balance calculation results for backbone portion of some protein main chains

No.	Species	Protein Name	PDB ID	Δ_x	Δ_y	Δ_z	r_x	r_y	r_z
1	Sperm whale	Myoglobin	1CQ2	2.19197	5.43389	-13.8403	1.0005	1.00083	0.994793
2	<i>Homo sapiens</i>	Plasminogen activator	1KDU	-5.96687	1.01917	-1.52315	0.998866	1.00116	0.999042
3	<i>Bacillus licheniformis</i>	KerA protein	6DWQ	-1.32691	-20.6017	-7.57049	0.999964	1.00073	0.998744
4	<i>Rhodobacter capsulatus</i>	Cytochrome C	1CPQ	10.1240	2.77415	-12.8370	1.00139	1.00046	0.998238

From Tables 1 and 2, it can be seen that from the perspective of force balance, the proteins in the tables can indeed be divided into two separately calculable parts, and the backbone and non-backbone portions of these protein main chains are respectively in a state of force balance. We hypothesize that force balance is an important characteristic of protein structure.

2.2 Protein Side Chain Portion

Using the method described in the theoretical section, we also calculated the force balance results for the non-backbone portion of some protein side chains, as shown in Table 3. The calculations include the force between the α -carbon and the β -carbon atoms as a part of protein side-chain non-backbone portion.

Table 3. Force balance calculation results for the non-backbone portion of some protein side chains

No.	Species	Protein Name	PDB ID	Δ_x	Δ_y	Δ_z	r_x	r_y	r_z
1	Sperm whale	Myoglobin	1CQ2	0.027297 2	3.6563	-3.31142	1.0000	1.00018	0.999569
2	<i>Homo sapiens</i>	Plasminogen activator	1KDU	-2.34295	-2.05589	1.24824	0.999841	0.999195	1.00026

3	<i>Bacillus</i> <i>licheniformis</i>	KerA protein	6DWQ	3.79525	-6.13377	9.55613	1.00008	1.00019	1.00117
4	Homo sapiens	Parathyroid hormone receptor	1BL1	-2.87593	0.514348	-2.70747	0.999407	1.00012	0.998956

From Table 3, it can be seen that the non-backbone portions of the side chains in the listed proteins also agree well with the force balance condition, indicating the significance of proposing this force balance expression. Force balance calculations for the backbone portion of protein side chains, particularly concerning the ring atoms in phenylalanine and tyrosine, as well as heterocyclic atoms in tryptophan and histidine, are still under investigation; therefore, relevant data are not provided in this paper.

From the calculation results in Tables 1 to 3, it is evident that although some central atoms in the main chain of the listed proteins may not individually satisfy the force balance condition, the backbone and non-backbone portions as a whole generally satisfy the force balance conditions well, allowing their force balance to be studied separately. The situation is similar for protein side chains. This so-called "independence" of the various parts of a protein in terms of force balance can greatly enhance the convenience of protein structure research.

It should be noted that calculations indicate not all protein main chains and side chains (both backbone and non-backbone portions) perfectly satisfy the force balance conditions. We believe that deviations from force balance may be related to factors such as bound small organic molecules, which will be discussed in detail in our future work.

Protein structure prediction is a significant problem in life sciences. Capturing the underlying principles of protein folding to achieve accurate prediction of protein structures has been one of the most important challenges in structural biology¹⁵. In this sense, the simple and precise formulas and principles related to the force balance of various parts of the protein main chain and side chains discovered in this work are undoubtedly beneficial.

Computer codes used in this paper are available at: <https://github.com/tongji92774>.

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